

Intelligence

Lecture 5

Intelligence & Life Outcomes

Dr Steven Poulter

Key Themes



- Does intelligence matter in school and the workplace?
- Is there a link between higher intelligence and health outcomes?

Glossary of Key Terms

Cognitive Ability Tests (CAT)

Standardised tests measuring verbal, non-verbal, and quantitative reasoning in UK schools. Shown to predict GCSE performance extremely well.

GCSE (General Certificate of Secondary Education)

National exams taken at age 16 in the UK. Used in the lecture to show how intelligence measured at age 11 predicts school outcomes.

Educational Attainment

The highest level of education an individual reaches (e.g., GCSEs, A-levels, university degree).

SES (Socio-economic Status)

A person's social and economic position, often measured via income, education, and occupation. Found to independently predict educational outcomes.

Personality Traits (Big Five)

Characteristics such as conscientiousness, neuroticism, extraversion, openness, agreeableness. Personality predicted social–emotional outcomes more strongly than education-related outcomes.

Meta-analysis

A statistical method that combines results from many studies to estimate overall effects. Used in the lecture (e.g., Schmidt & Hunter's job-performance research).

Job Performance

How well someone performs their occupational role. Often measured using supervisor ratings or work samples. Intelligence is one predictor, though the evidence base has limitations.

Occupational Sorting

The tendency for people with different levels of cognitive ability to enter different types of jobs. Large UK datasets show systematic differences in mean cognitive ability between occupations.

Cultural Capital

Long-term knowledge, skills, and behaviours acquired through education and experience. Proposed as an indirect link between intelligence and later job performance.

Scottish Mental Surveys (1932 & 1947)

Nationwide IQ-type testing of almost all 11-year-olds in Scotland. Enabled uniquely powerful research linking childhood intelligence with later-life health and longevity.

Longitudinal Study

A study following the same individuals over long periods. Used to evaluate intelligence stability and long-term outcomes.

Cognitive Ageing

Changes in cognitive ability across the lifespan. Intelligence measured at 11 strongly predicts intelligence decades later (correlation ~ 0.63 – 0.70).

System Integrity Hypothesis

Idea that intelligence reflects overall biological ‘wiring’ quality of the body, possibly linking higher intelligence to better health and longevity.

APOE Gene

A genetic marker associated with cognitive decline and Alzheimer’s risk. Identified in systematic reviews of risk factors for cognitive ageing.

Mortality Risk

Likelihood of death across a specific timeframe. Childhood intelligence has a robust negative association with mortality risk (higher IQ $\rightarrow$ lower risk).

Health Behaviours

Lifestyle choices (e.g., smoking, diet, exercise) that influence health. Higher childhood intelligence is associated with healthier behaviour patterns by adulthood.

Pre-morbid Intelligence

Intelligence measured before illness onset. Important for interpreting studies showing IQ predicts later health conditions.

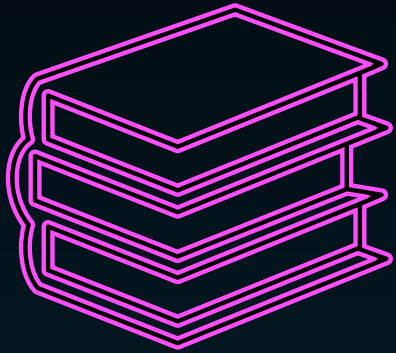
Mediterranean Diet

A dietary pattern identified as protective for cognitive ageing (more vegetables, olive oil, fish, whole grains).

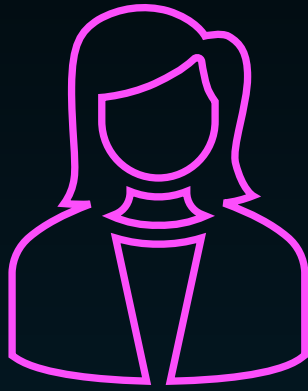
Deprivation (Childhood SES)

Childhood disadvantage linked to many outcomes but only partially explains intelligence–health associations.

Applications of intelligence tests



Education



Workplace



Health



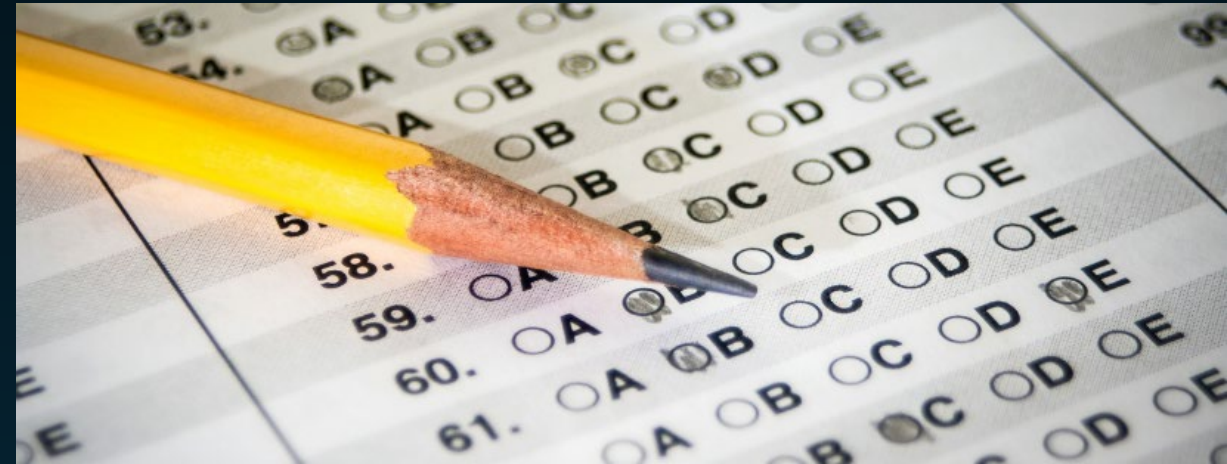
Intelligence & educational achievement

Do intelligence test scores predict educational success?

- 1944: In UK, intelligence tests widely used to select children at 11+ years into an educational 'strand' (*academic, technical or functional*)
- Idea was that skills were more important than a pupil's financial resources and that different skills required different schooling
- Results of the tests used to match children's abilities to their schools
- The reality was that the required specialist schools were not available at the scale required so the 11+ came to be a fierce competition for places at desirable grammar schools
- The passing or failing took on extra emphasis and become resented by many (was phased out by most of UK by 1976)
- Intention was for the 11+ to be like an IQ test consisting of 3 papers: *Arithmetic, Writing, General Problem Solving* (sometimes with an added *Verbal* paper). Most tests multiple choice (40-60 minutes long)

Do intelligence test scores predict educational success?

- 1970s onwards the UK school leaving age raised to 16 years and education became mostly 'comprehensive' (not technically selecting pupils on academic aptitude)
- However, under another name and for other purposes, mass cognitive testing still takes place in UK schools
- These are called 'cognitive ability tests' (CAT) rather than intelligence tests
- Used for estimating student progress within the context of how well a school is performing



What does psychometric intelligence at 11 years predict?



The Cognitive Abilities Test
(CAT)

11 years old

Good predictor?



General Certificate of Secondary
Education (GCSE)

16 years old

What does psychometric intelligence at 11 years predict?

2007 Intelligence and educational achievement

Ian J. Deary ^{a,*}, Steve Strand ^b, Pauline Smith ^c, Cres Fernandes ^c

Comparison of the matched sample against the population average for GCSE/GNVQ outcomes

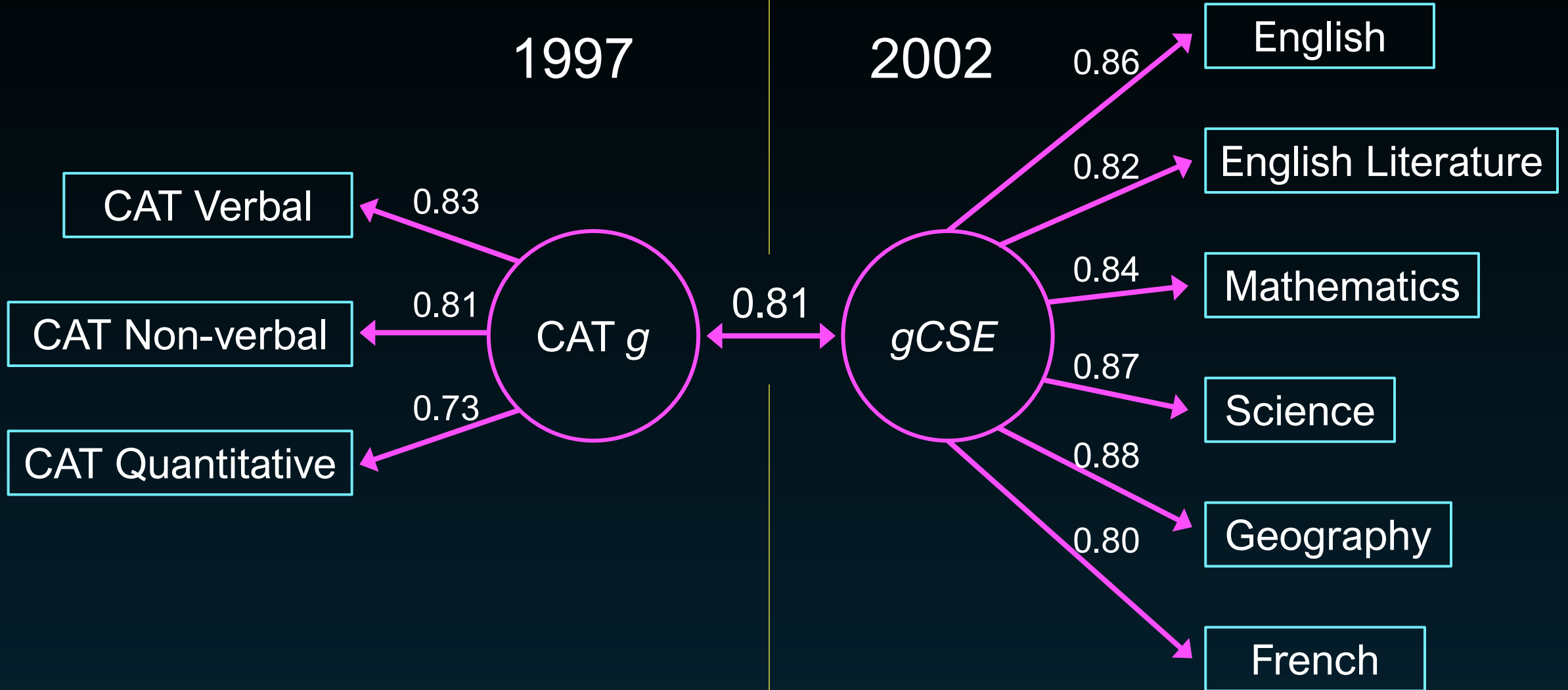
Measure	CAT-matched sample	Full national dataset
GCSE/GNVQ 'Best 8' points: mean (S.D.)	37.3 (13.4)	36.8 (14.0)
English: % entries graded A*–C	62%	61%
Maths: % entries graded A*–C	54%	53%
Science: % entries graded A*–C	54%	53%
Sample size	80,074	361,335

- 74,403 children (37,509 girls, 36,894 boys)
- 973 secondary schools across 103 LEAs (1/5 of total national dataset)
- CAT test scores 1997 vs GCSEs 2002



11 years old

16 years old



Adapted from Deary (2020)

Correlations between general intelligence (CAT *g*) and GCSE scores

Cognitive Abilities Test	
<i>g</i> factor ^b	–
Residual verbal factor ^b	0.00
Overall score	
GCSE total points	0.69 (70,530)
GCSE Best 8	0.72 (68,904)
Arts and Humanities	
English	0.67 (67,677)
English literature	0.59 (62,416)
Drama	0.47 (10,997)
Religious Education	0.52 (13,572)
French	0.64 (36,370)
German	0.61 (16,638)
Spanish	0.62 (6501)
Science	
Mathematics	0.77 (68,125)
Double Science	0.68 (59,518)
Single Science	0.60 (5331)
Physics	0.50 (2733)
Chemistry	0.46 (2720)
Biology	0.51 (2764)
Social Sciences	
Geography	0.65 (26,081)
History	0.63 (22,764)
Business	0.56 (11,188)
Information Technology	0.47 (9350)
Information Technology short course	0.48 (8931)
Practical	
Art and Design	0.43 (15,104)
Music	0.54 (5208)
Physical Education	0.55 (13,846)
DT-Food	0.52 (13,493)
DT-Graphics	0.45 (14,328)
DT-Resistant Materials	0.48 (14,059)
DT-Textiles	0.52 (6390)

Table 2
Sex comparisons in cognitive ability and GCSE scores, and correlations between general cognitive ability (*g*) and the residual verbal factor and GCSE scores

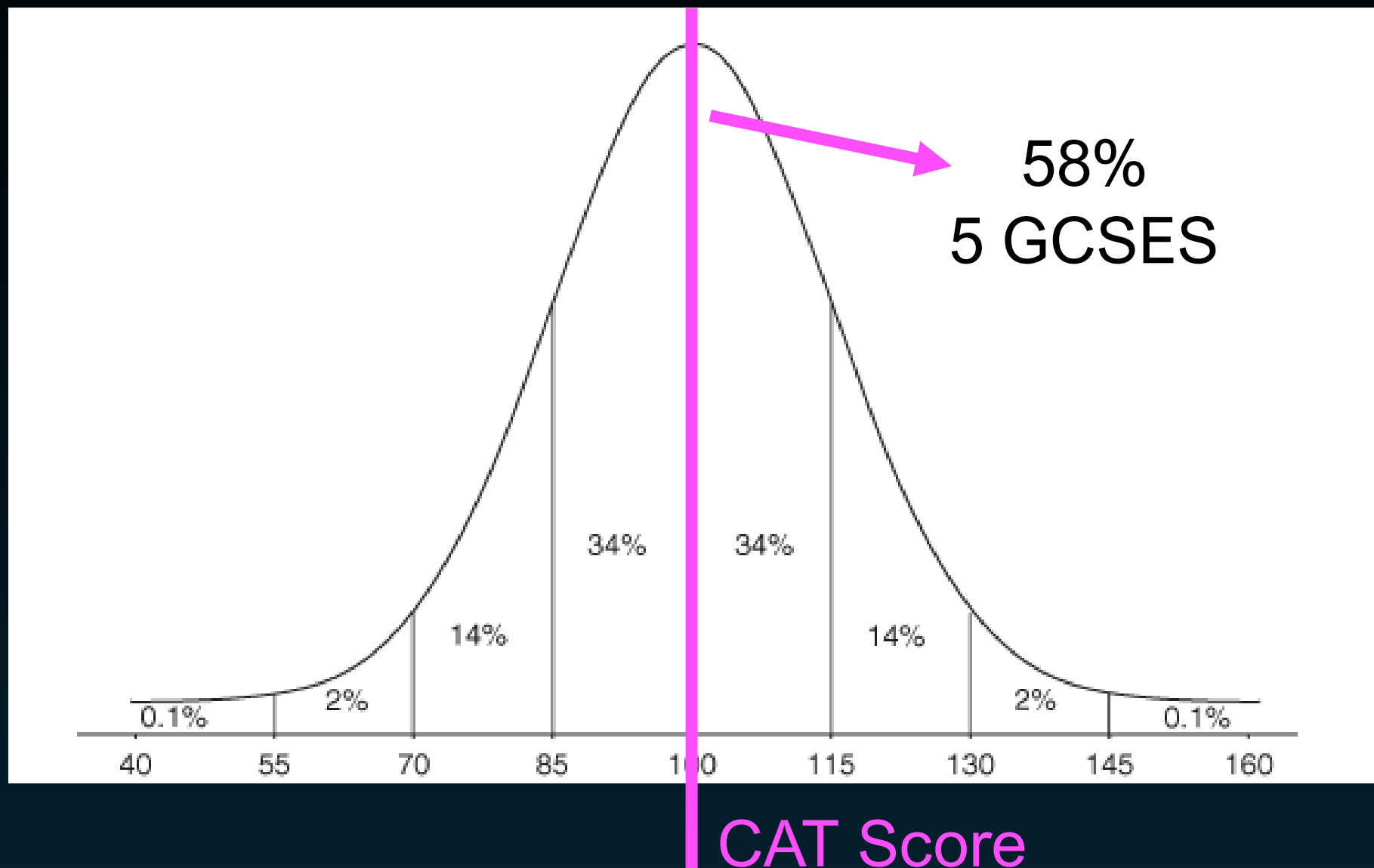
Cognitive Ability Test or GCSE subject	Male–female comparisons			Correlations	
	Boys’ mean (S.D., <i>N</i>)	Girls’ mean (S.D., <i>N</i>)	<i>p</i> for difference (Cohen’s <i>d</i>)	CAT <i>g</i>	CAT residual verbal ^a
Cognitive Abilities Test					
<i>g</i> factor ^b	–0.007 (0.965, 34,850)	0.004 (0.912, 35,680)	0.096 (0.01)	–	
Residual verbal factor ^b	–0.126 (1.026, 34,850)	0.122 (0.958, 35,680)	<0.001 (0.25)	0.00	–
Overall score					
GCSE total points	39.5 (18.8, 36,894)	45.1 (18.2, 37,509)	<0.001 (0.30)	0.69 (70,530)	0.13
GCSE Best 8	35.4 (13.5, 35,848)	39.4 (12.8, 36,759)	<0.001 (0.30)	0.72 (68,904)	0.14
Arts and Humanities					
English	4.52 (1.53, 34,947)	5.13 (1.41, 36,328)	<0.001 (0.41)	0.67 (67,677)	0.22
English literature	4.60 (1.61, 31,316)	5.22 (1.44, 34,317)	<0.001 (0.41)	0.59 (62,416)	0.20
Drama	4.97 (1.52, 4022)	5.53 (1.38, 7537)	<0.001 (0.39)	0.47 (10,997)	0.14
Religious Education	4.16 (2.07, 5887)	4.96 (1.93, 8211)	<0.001 (0.40)	0.52 (13,572)	0.16
French	4.04 (1.69, 17,876)	4.74 (1.69, 20,213)	<0.001 (0.41)	0.64 (36,370)	0.20
German	4.28 (1.62, 8135)	4.93 (1.60, 9491)	<0.001 (0.40)	0.61 (16,638)	0.18
Spanish	3.92 (1.82, 2734)	4.71 (1.76, 3983)	<0.001 (0.44)	0.62 (6501)	0.18
Science					
Mathematics	4.48 (1.73, 35,371)	4.54 (1.71, 36,390)	<0.001 (0.03)	0.77 (68,125)	0.00
Double Science	4.52 (1.60, 30,831)	4.63 (1.61, 31,918)	<0.001 (0.07)	0.68 (59,518)	0.12
Single Science	2.95 (1.45, 2661)	3.30 (1.48, 2928)	<0.001 (0.24)	0.60 (5331)	0.02
Physics	5.83 (1.35, 1555)	5.78 (1.39, 1268)	0.33 (–0.04)	0.50 (2733)	0.09
Chemistry	5.61 (1.33, 1539)	5.92 (1.26, 1272)	<0.001 (0.24)	0.46 (2720)	0.08
Biology	5.84 (1.22, 1568)	6.07 (1.21, 1292)	<0.001 (0.19)	0.51 (2764)	0.14
Social Sciences					
Geography	4.55 (1.77, 15,014)	4.88 (1.80, 12,430)	<0.001 (0.18)	0.65 (26,081)	0.16
History	4.62 (1.95, 11,697)	5.02 (1.91, 12,220)	<0.001 (0.21)	0.63 (22,764)	0.18
Business	4.49 (1.77, 6591)	4.72 (1.74, 5184)	<0.001 (0.13)	0.56 (11,188)	0.11
Information Technology	4.43 (1.87, 5747)	4.79 (1.83, 3840)	<0.001 (0.19)	0.47 (9350)	0.07
Information Technology short course	3.79 (1.88, 4144)	4.39 (1.90, 5217)	<0.001 (0.32)	0.48 (8931)	0.11
Practical					
Art and Design	4.59 (1.62, 6486)	5.54 (1.49, 9397)	<0.001 (0.61)	0.43 (15,104)	0.09
Music	5.11 (1.97, 2221)	5.48 (1.70, 3224)	<0.001 (0.20)	0.54 (5208)	0.16
Physical Education	4.87 (1.53, 9802)	5.03 (1.65, 4696)	<0.001 (0.10)	0.55 (13,846)	0.07
DT-Food	4.14 (1.68, 3605)	4.84 (1.65, 10,626)	<0.001 (0.42)	0.52 (13,493)	0.08
DT-Graphics	4.24 (1.80, 8387)	5.06 (1.67, 6745)	<0.001 (0.47)	0.45 (14,328)	0.07
DT-Resistant Materials	4.18 (1.67, 11,361)	4.88 (1.69, 3499)	<0.001 (0.42)	0.48 (14,059)	0.06
DT-Textiles	3.68 (1.86, 201)	5.00 (1.65, 6557)	<0.001 (0.75)	0.52 (6390)	0.09

- There were significant sex differences ($p<0.001$) in virtually all GCSE scores
- Girls performed better in every topic except Physics. The effect sizes of the sex differences were often substantial (Art & Design had a Cohen’s *d* of 0.61!!)
- Girls and boys did not differ in *g* but girls scored higher on residual verbal factor

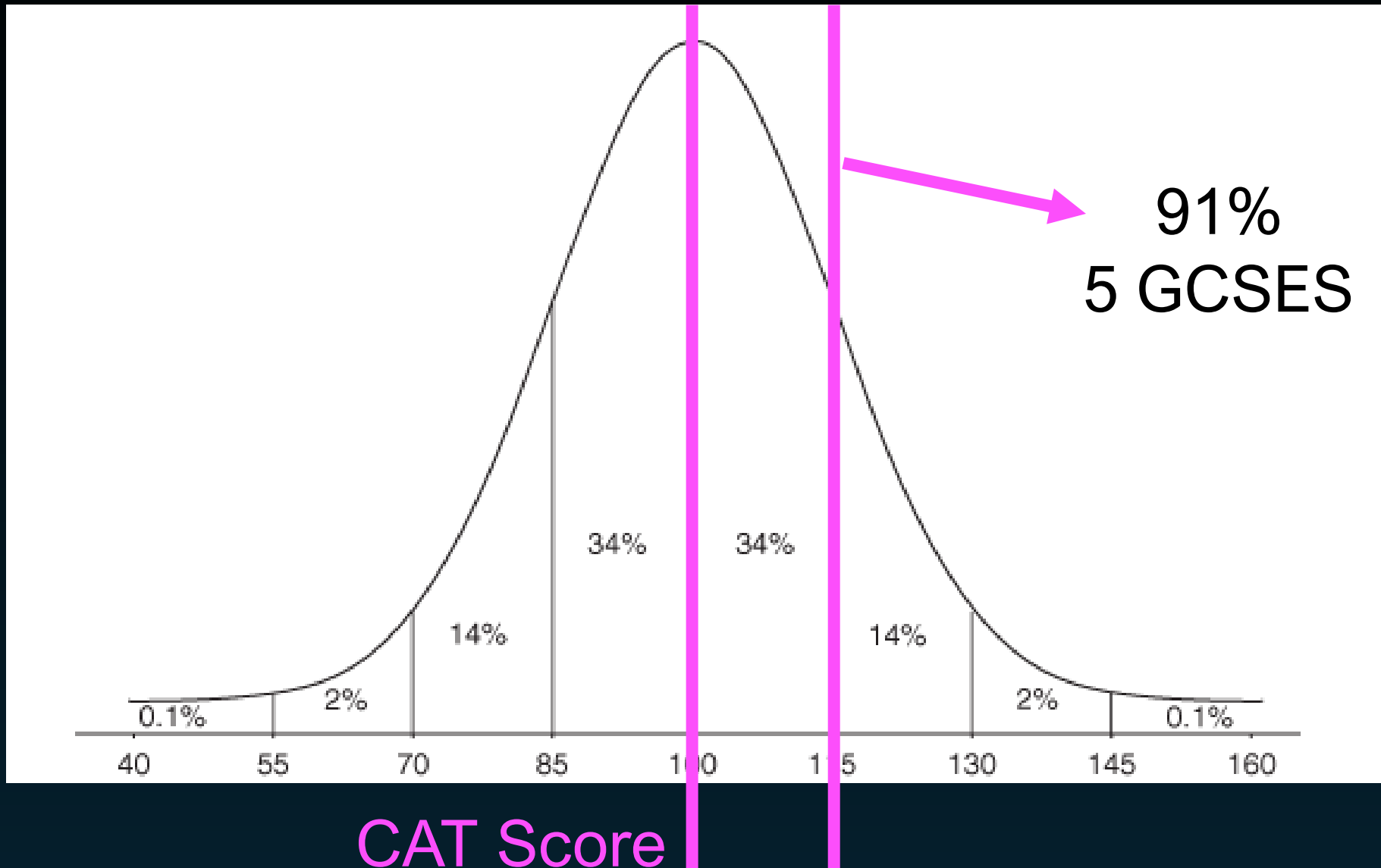
Quick Summary

- Correlation between general intelligence at age 11 and general GCSE performance at age 16 years was **0.81** – this is high!
- People's intelligence differences at age 11 are a powerful predictor of their differences in educational outcomes at age 16.
- Educational outcomes are doorways to higher education, jobs and professions. Thus, important people's intelligence test scores can be optimised by age 11.

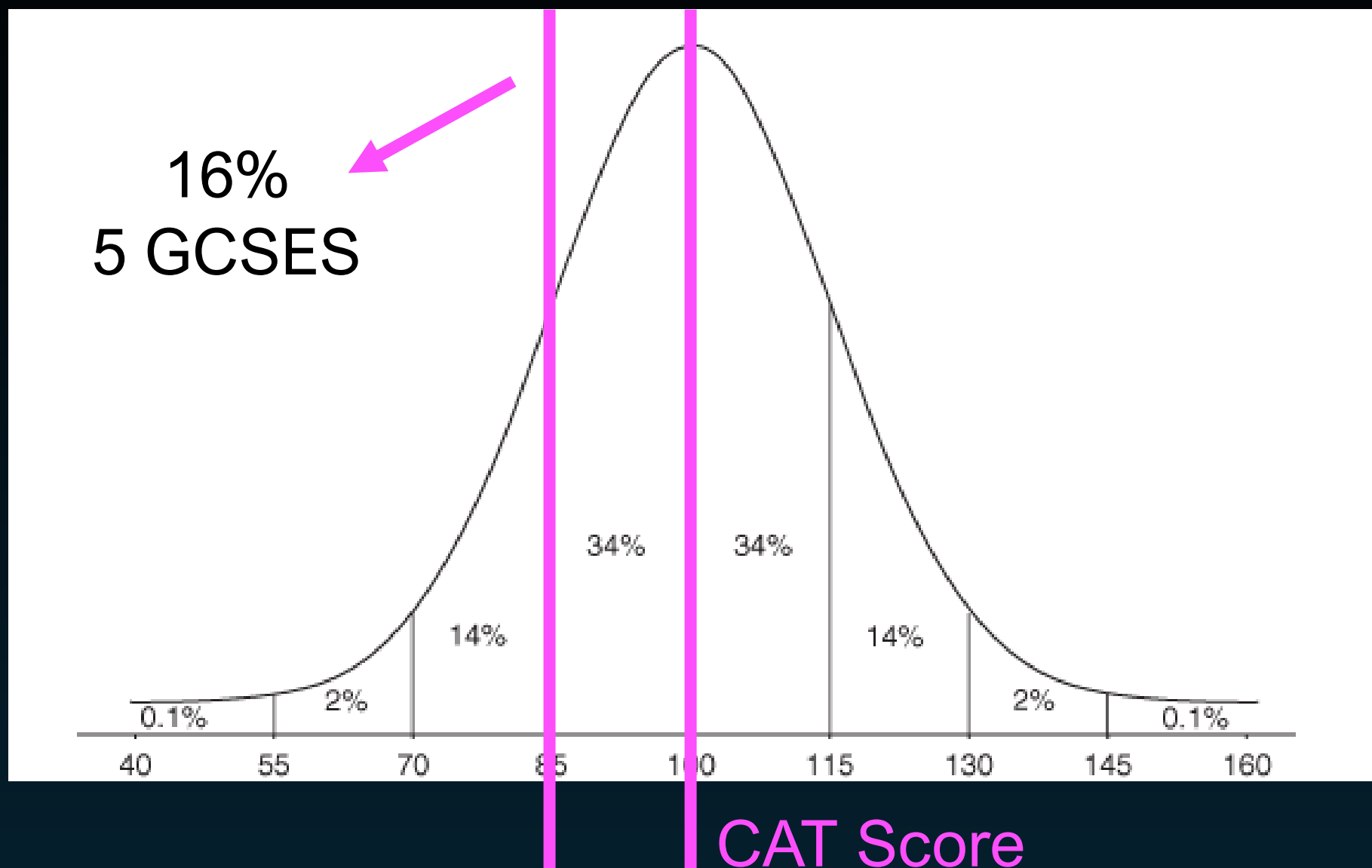
Association between CAT and gaining 5 or more GCSEs at A* to C



Association between CAT and gaining 5 or more GCSEs at A* to C

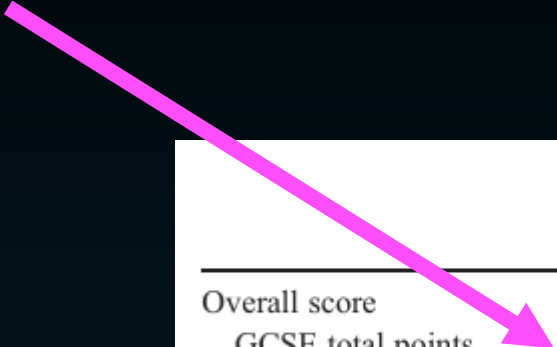


Association between CAT and gaining 5 or more GCSEs at A* to C



Quick Summary

- Strong results for intelligence as a **predictor**
- **However**, the correlations indicate that around half of the variance in GCSE scores can be statistically explained by the prior *g* factor at 11 years old
- Thus, a large proportion of variance needs to be explained by non-*g* factors, which may include school attendance/engagement, personality traits, motivation and effort, parental support, teaching quality, school ethos etc. **Intelligence not close to being the whole story but a substantial part of educational success in this context.**



	CAT <i>g</i>
Overall score	
GCSE total points	0.492
GCSE Best 8	0.535

GCSE grades a good predictor of life chances and wellbeing, research shows

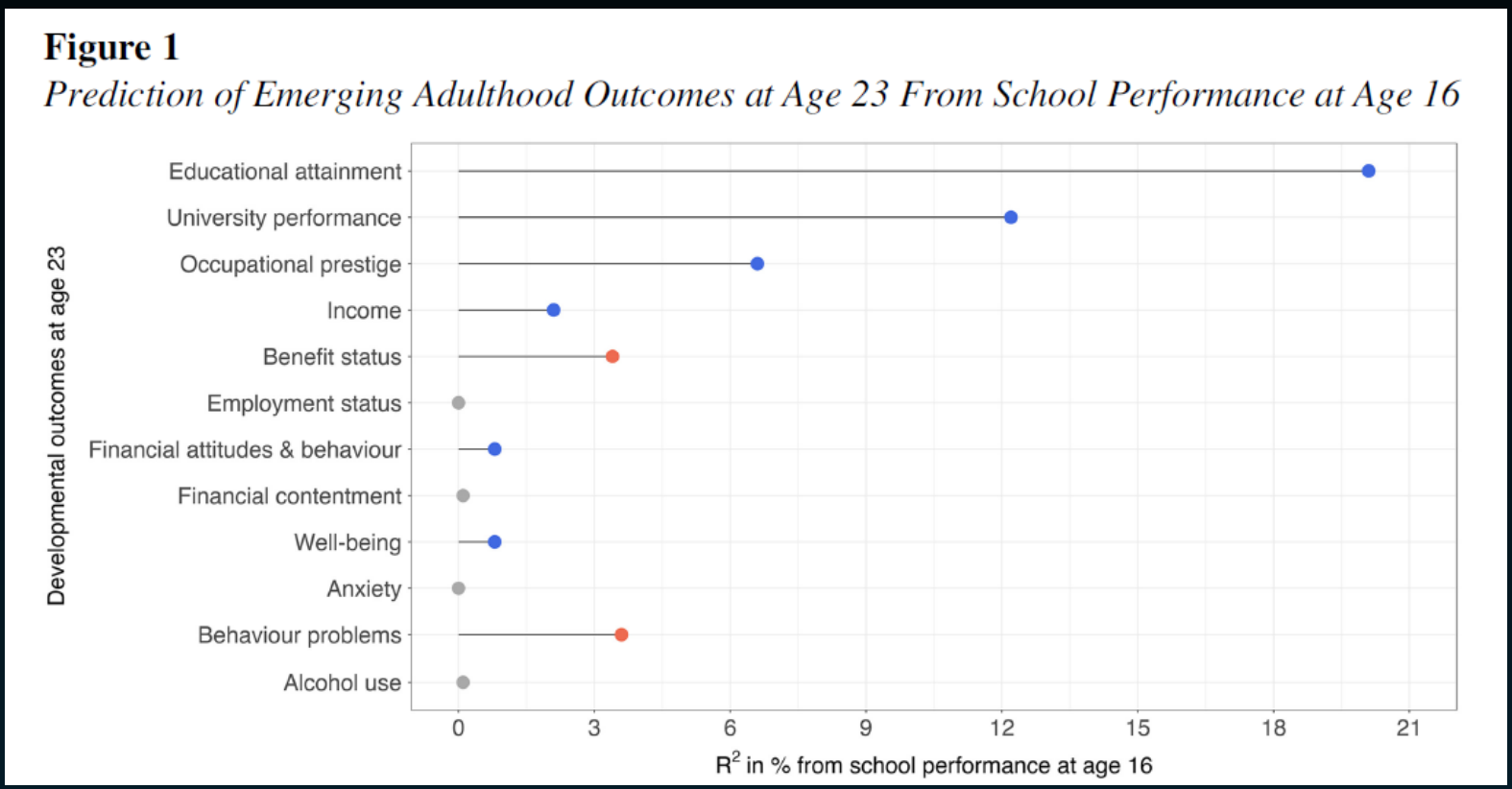
Study of 23-year-olds found the exams were even more important for those from disadvantaged backgrounds



The study found the predictive power of GCSEs outstripped gender or going to university. Photograph: David Davies/PA

GCSE grades have an excellent track record in predicting the future lives and careers of young adults, according to researchers, who found the exams were even more crucial for those from disadvantaged backgrounds.

Academics from Leeds and York universities found that the predictive power of GCSE results outstripped those of gender or later qualifications, including university degrees, in charting the development of young people from the age of 16 into their early 20s.

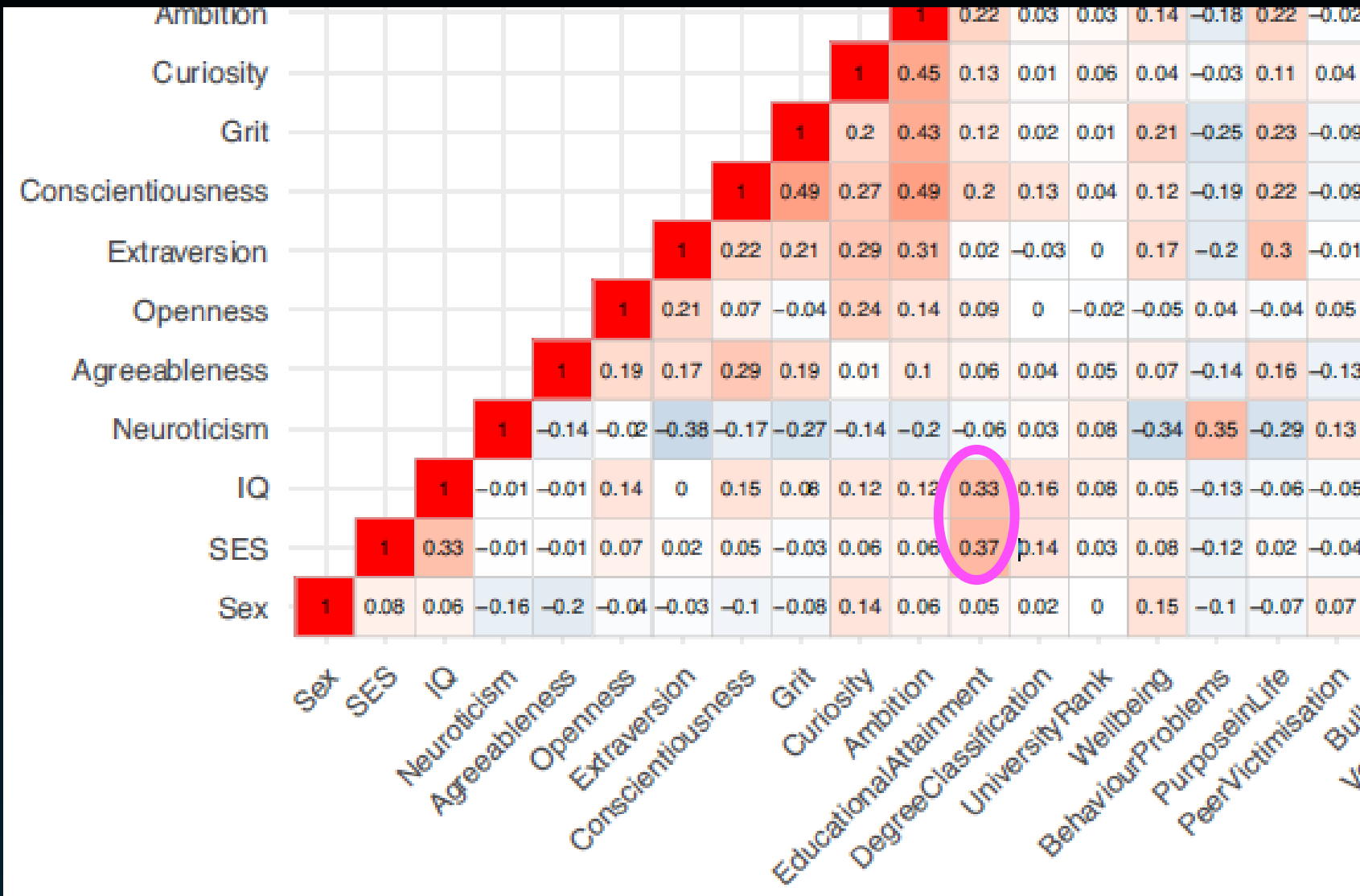


Predicting Educational and Social–Emotional Outcomes in Emerging Adulthood From Intelligence, Personality, and Socioeconomic Status

Zainab Faatimah Haider and Sophie von Stumm
Department of Education, University of York

- Tested intelligence and personality in adolescence to see if scores could predict educational outcomes in emerging adulthood
- $n = 2,277$. Tested at 16 years (Raven's Matrices and Mill Hill vocabulary scale) and also on different measures of personality

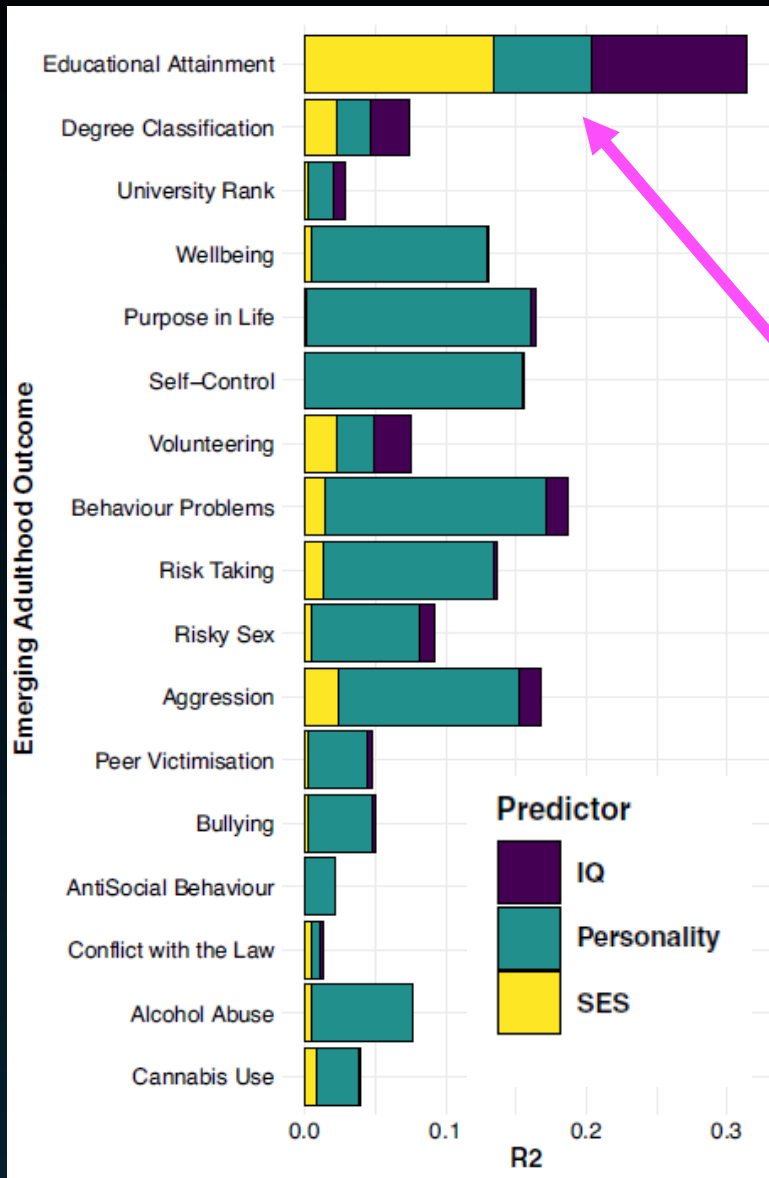
Intelligence & educational attainment



Intelligence and SES more predictive of educational outcomes (highest qualification)

Personality stronger predictor of social-emotional outcomes

Intelligence & educational achievement



What about individual contributions to educational attainment?

- Statistical model to separate individual contributions of each outcome
- About a quarter of variance in educational outcomes explained by IQ, personality and SES combined.
- IQ scores more predictive for educational qualifications than for other outcomes measured (e.g. conflict with the law)

Psychological Correlates of University Students' Academic Performance: A Systematic Review and Meta-Analysis

Michelle Richardson
University College London

Charles Abraham
University of Exeter

Rod Bond
University of Sussex

- IQ scores predict university GPA – $r = 0.20$
- Do remember this is only a sub-population of people (self-selecting, higher IQ on average)



Intelligence & the workplace

Do intelligence test scores predict who will perform well in the workplace?

The Validity and Utility of Selection Methods in Personnel Psychology:
Practical and Theoretical Implications of 85 Years of Research Findings

Frank L. Schmidt
University of Iowa

John E. Hunter
Michigan State University

1998

- Meta-analysed studies on decision-making in job hiring
- Looked at studies conducted over 85 years of research
- Examined the predictive power of 19 different ways of selecting people for jobs



Do intelligence test scores predict who will perform well in the workplace?

- Schmidt and Hunter (2004) based on further meta-analysis went hardline on the high intelligence-job performance correlation:

‘Nearly 100 years ago, Spearman (1904) proposed that the construct of GMA (general mental ability) is central to human affairs. The research presented in this article supports his proposal in the world of work, an area of life critical to individuals, organizations, and the economy as a whole.’

- However, this work and view has been criticised for the unreliability of the statistics, measures of job performance (supervisor ratings), intelligence tests used, secondary reports and populations tested (see Richardson and Norgate, 2015 for review)

Relationship between intelligence and occupation

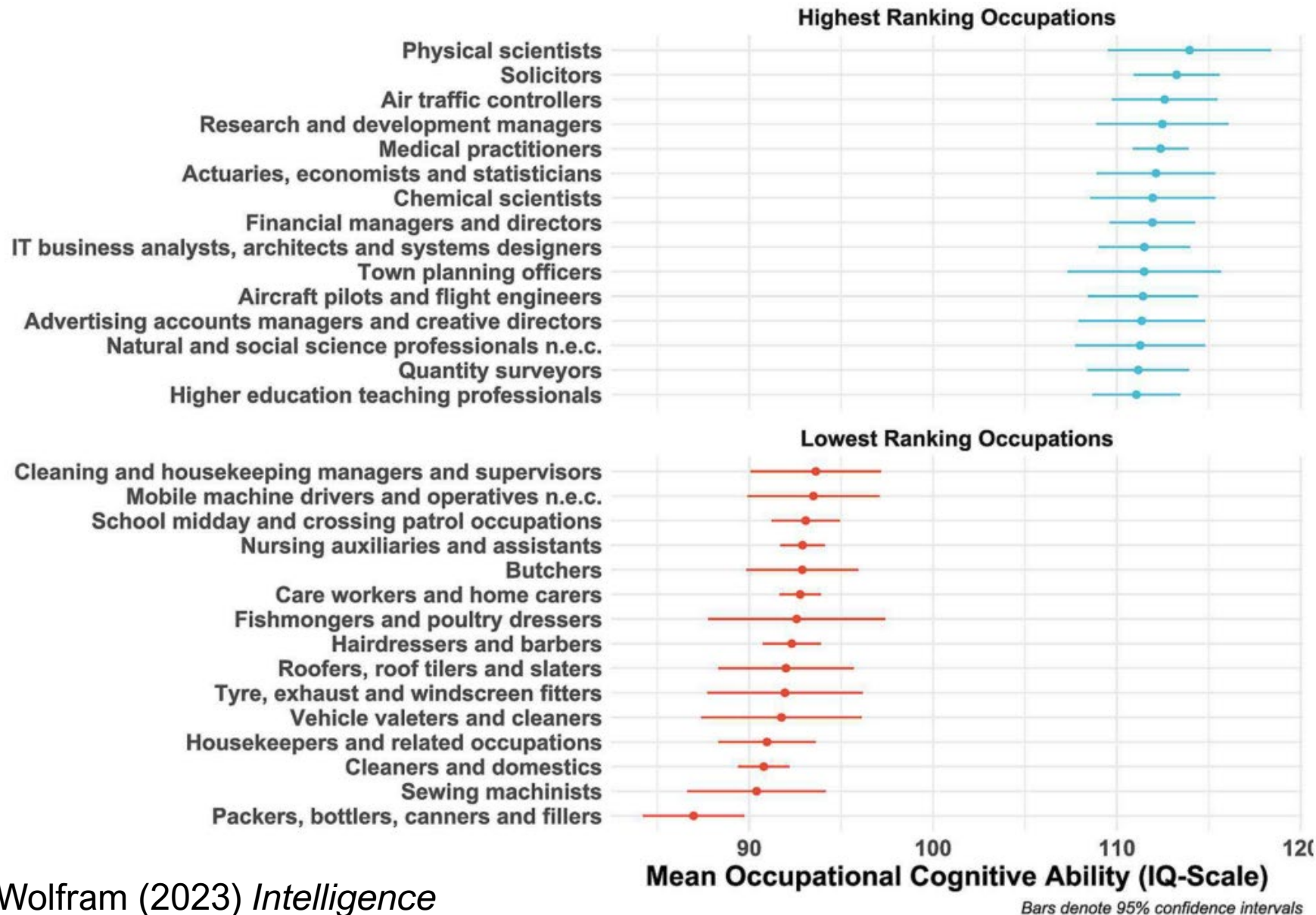
- *Psychometric view* – emphasises the role cognitive ability plays in occupational sorting
- Much criticism of this view based on lack of representative samples (lots of military data) and poor quality data

Table 1.1

Share of variance in cognitive ability associated with occupation in the literature.

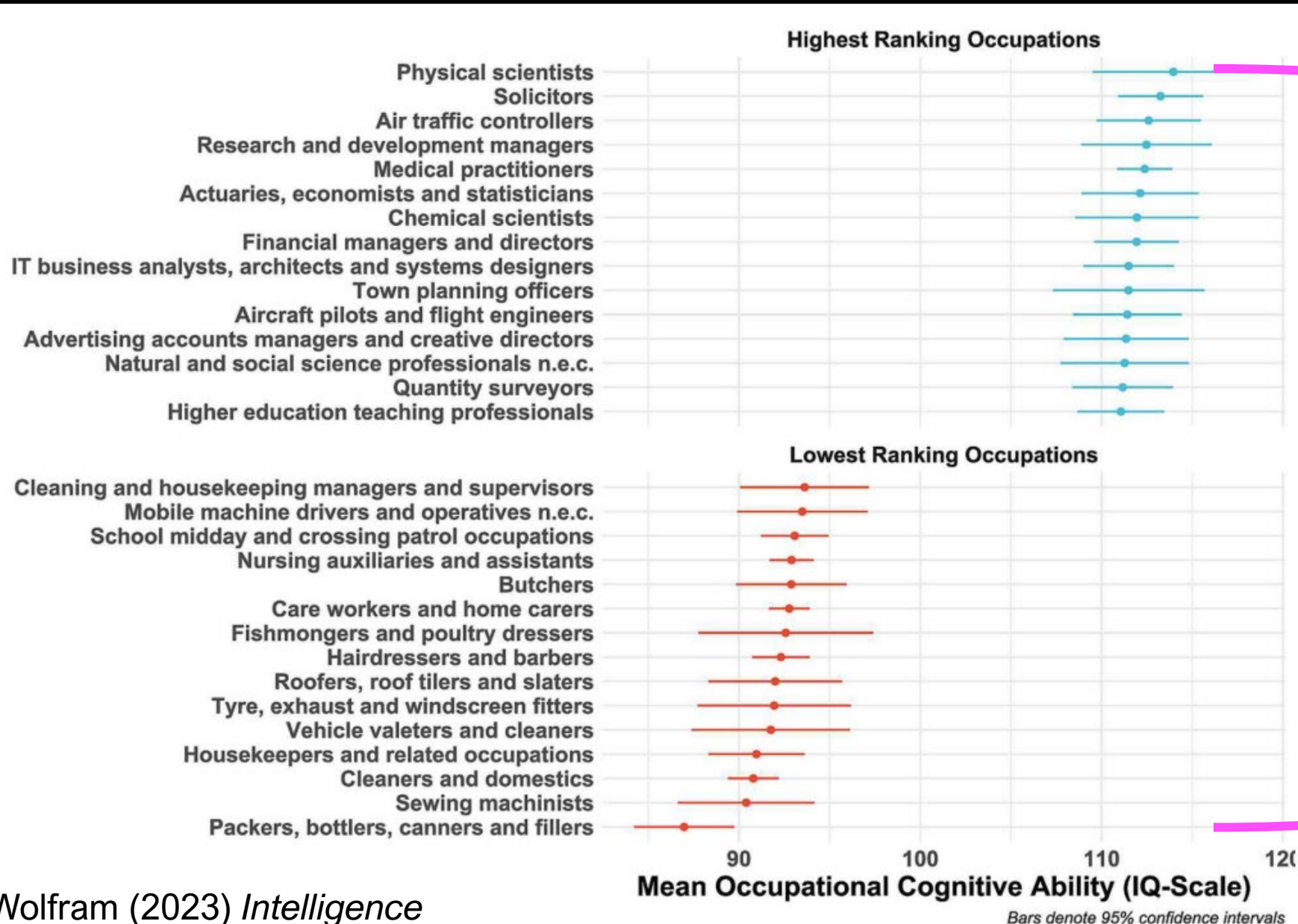
Publication	Population	Share
Stewart (1947)	White enlisted US men (WW2)	0.51
Jensen (1980)	US workforce, collected from late 40s to late 60s	0.47
Gottfredson (1997)	Job Applicants (US, 1983–1992)	0.31
Hauser (2010)	1957 Wisconsin Highschool Graduates (US, first job)	0.22
Hauser (2010)	1957 Wisconsin Highschool Graduates (US, 1975–1977 job)	0.20
Hauser (2010)	1957 Wisconsin Highschool Graduates (US, 1992–1993 job)	0.19
Hunt and Madhyastha (2012)	Job Applicants (US, unclear timeframe)	0.31
Huang (2013)	Extension of Gottfredson's (1997) applicant data (US, 1983–2002)	0.24
Huang (2013)	Repeated observations of NLSY79 panelists (US, 1983–2002)	0.20
Huang (2013)	Repeated observations of WLS panelists (US, first job to 1993)	0.18

Relationship between intelligence and occupation



- Analysis based on longitudinal survey of more than 40,000 UK households
- Occupation measures for ~56,000 respondents
- All 16+ respondents took 5 cognitive ability tests (numerical ability, working memory, problem solving, verbal semantic fluency, fluid reasoning)

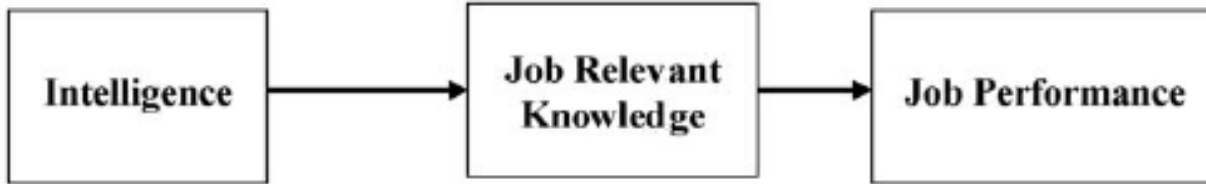
Relationship between intelligence and occupation



difference between the highest and lowest mean intelligence estimates amounts to almost two standard deviations (physical scientists, with 114 vs. packers, bottlers, canners and fillers, with 87)

Nature of relationship between IQ & Job Performance?

The Conventional Industrial Psychology Model*



*Adapted from Hunter, 1986.

Direct (short-term) relationship

A Sociological Alternative

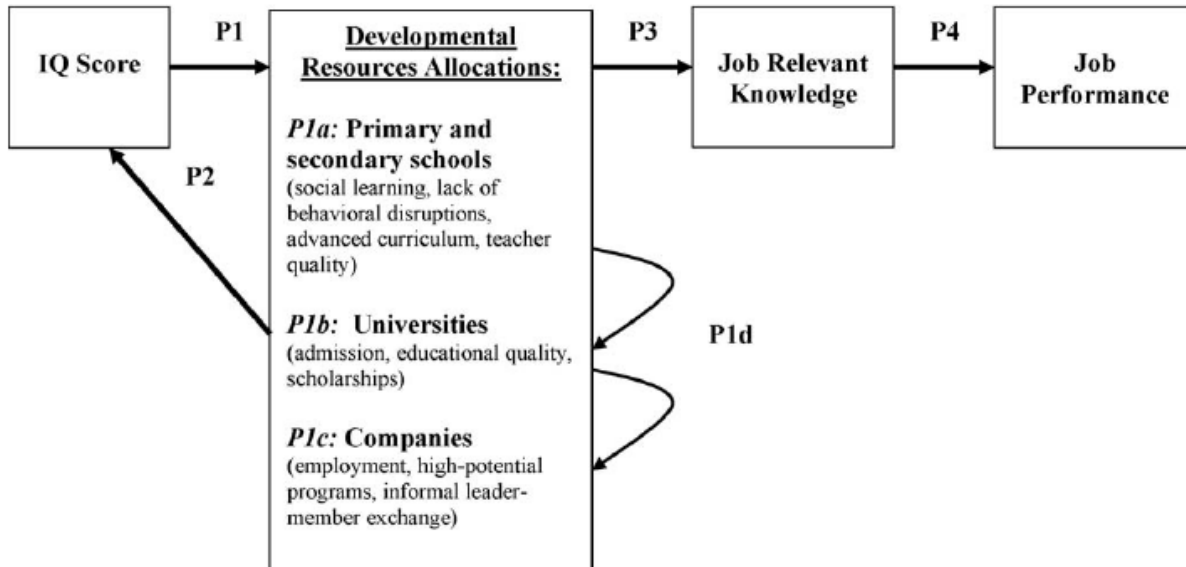


Fig. 2. A sociological alternative.

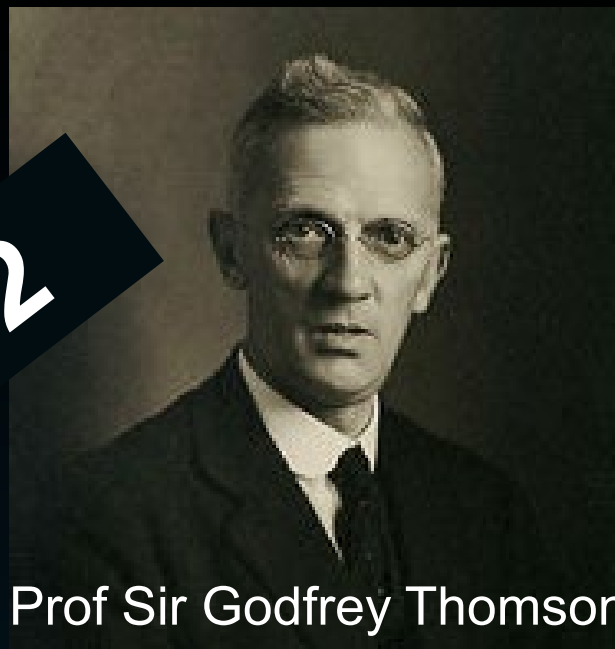
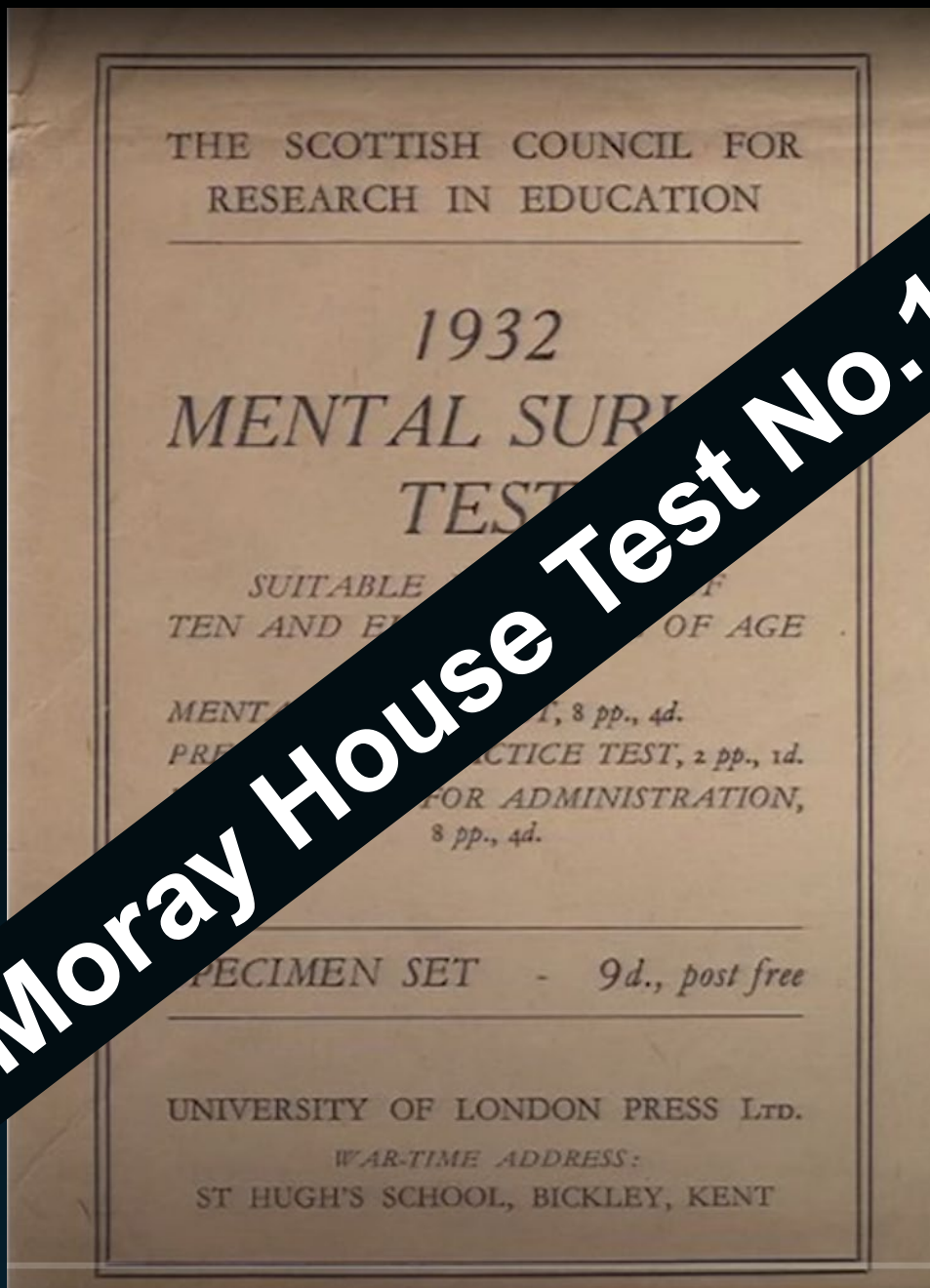
Indirect (long-term) acquisition of cultural capital

Quick Summary

- Some evidence for a link between higher IQ and better job performance
- However, vast bulk of evidence is based on meta-analyses combining studies of variable quality.
- A large study shows cognitive ability tests do quite a good job of occupational sorting, but note this is not predicting future job success based on prior intelligence levels.
- A more nuanced approach (considering indirect IQ-job performance links) from IQ-centric decision-making needs to be considered

Intelligence & health

Moray House Test No.12



Prof Sir Godfrey Thomson

Went to Armstrong College then of Durham University (1906 to 1925)

- Pioneer, who led rest of world, in standardisation of mental tests

- Obtained only record of IQ-type scores from full national year-of-birth cohorts (Scotland 1932 & 1947)
- By 1948 1.25 million Moray House Tests sold
- Advocate of universal education – he saw his mental ability tests as a useful means to give poor children a chance in life by looking past pupils' social status

1. Look at this example:-

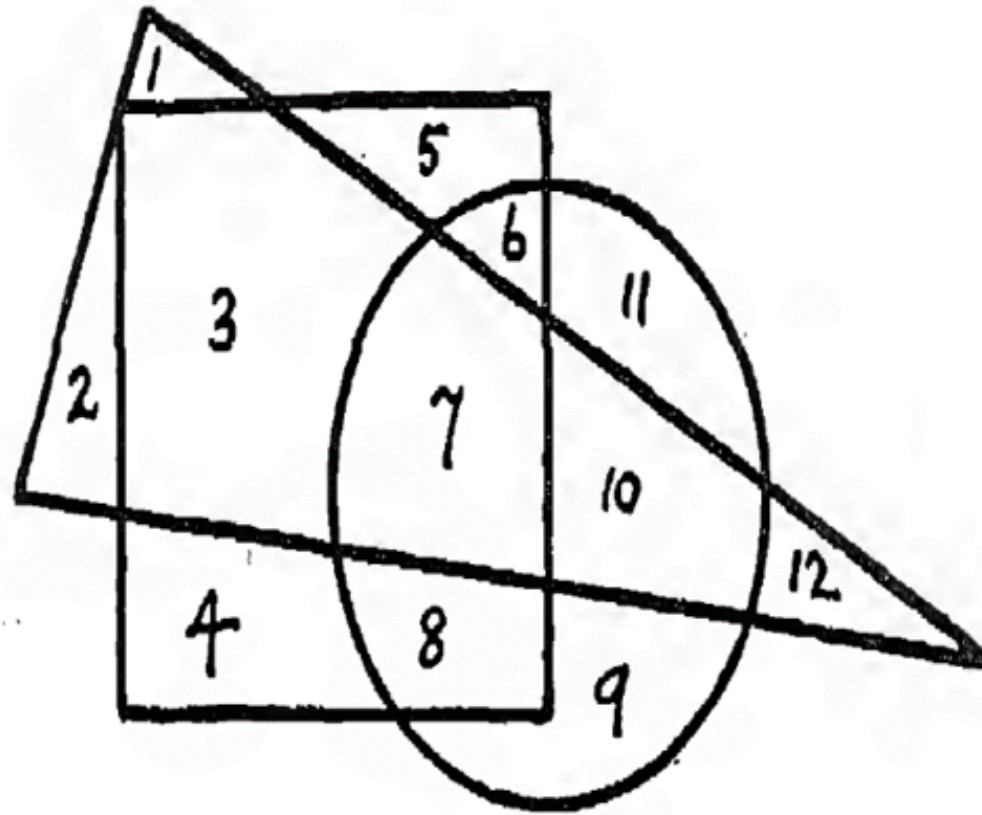
Finger is to toe as hand is to what? (foot, knee, arm, shoe, nail)

The answer is one of the words in the bracket. The right answer is foot, and it is underlined.

Now look at this next example, and then underline the right answer in brackets:

Foot is to man as hoof is to (leather, hard, cow, leg, boot)

10. What number is in the triangle and square but not in the circle?



Scottish Mental Surveys 1932 and 1947

- Ian Deary and colleagues traced survivors of the 1932 survey (now 77 years old)
- Exactly 66 years to the day that they took the first test
- Were able to obtain scores on same intelligence test at age 11 and 77 years
- How similar would people's relative scores be across most of their life span?



The all-important young baseline

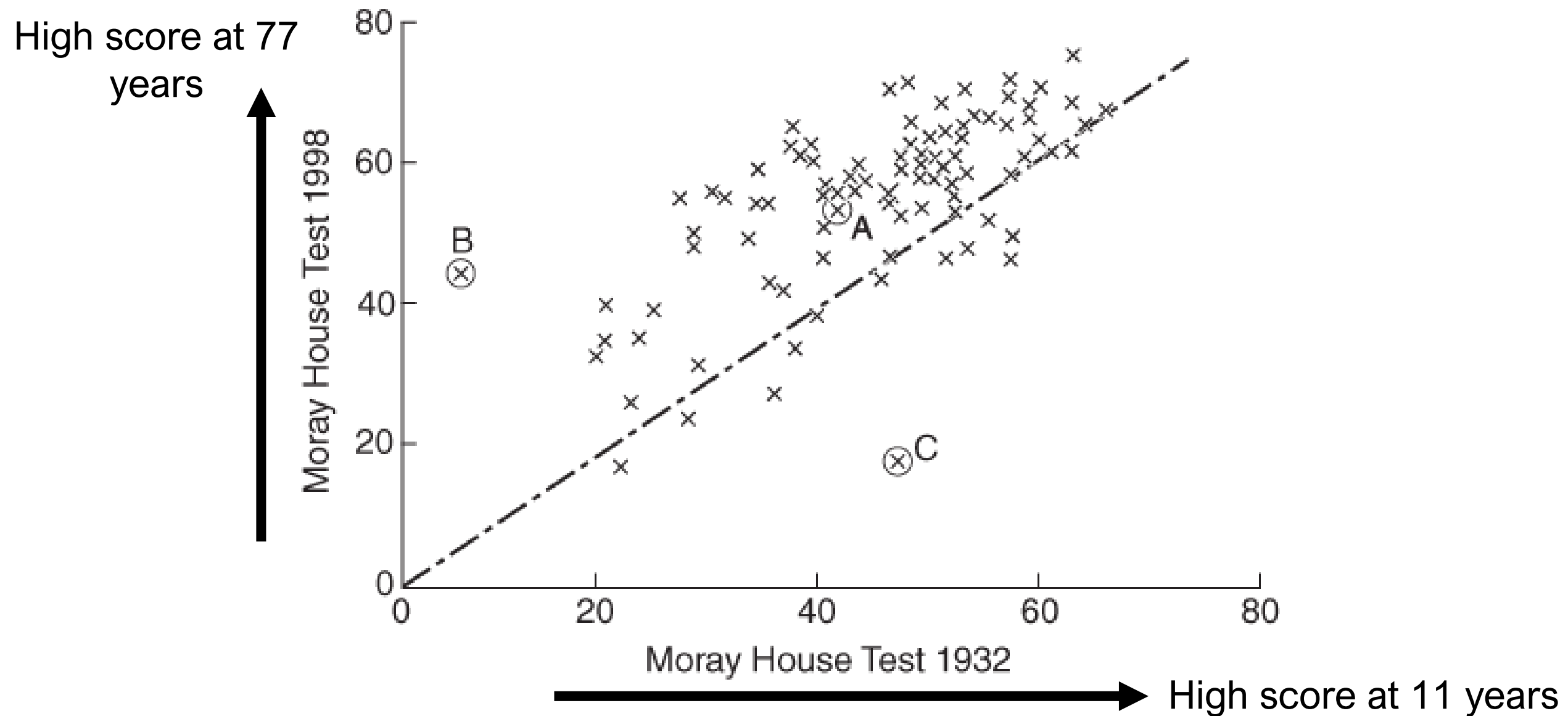


11 years old

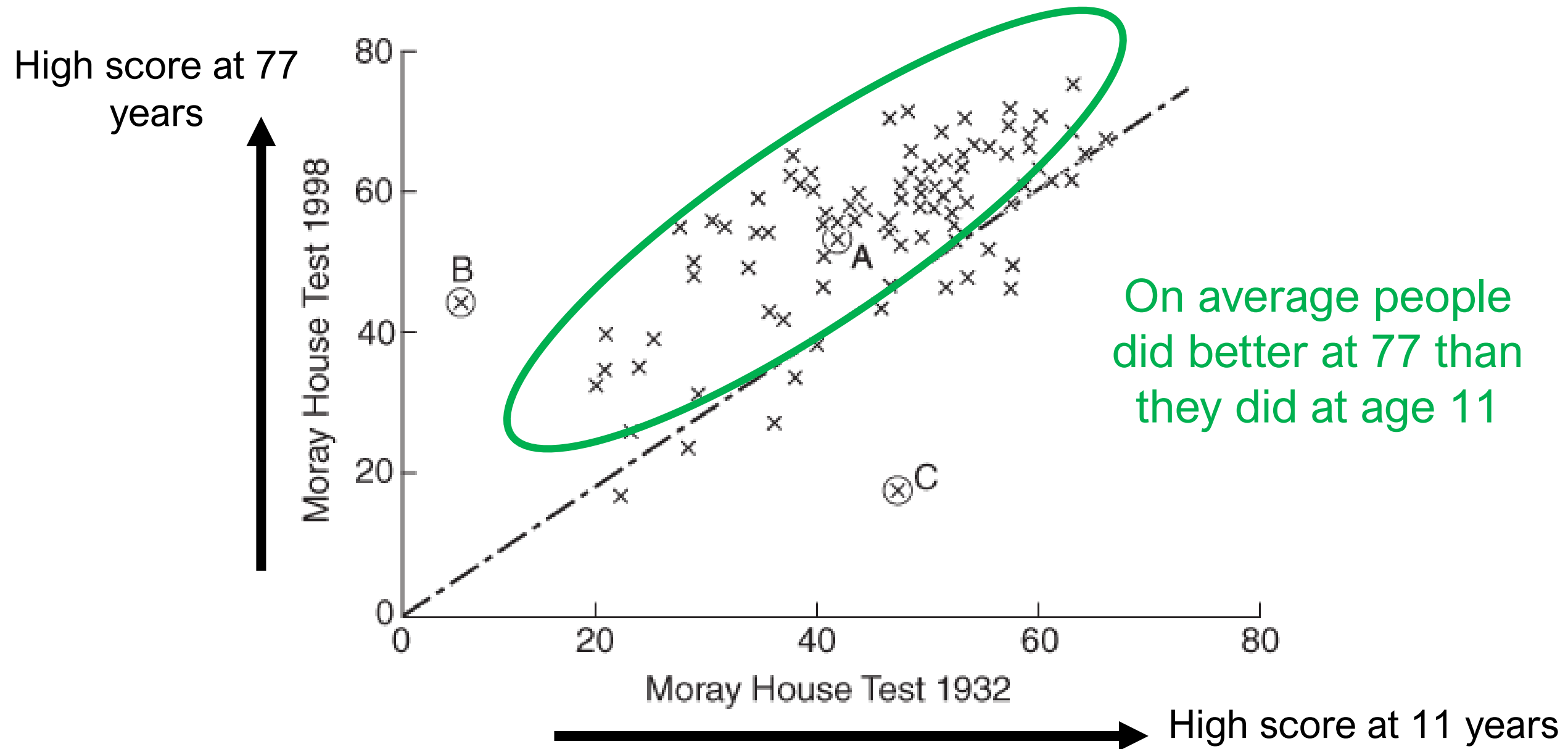


77 years old

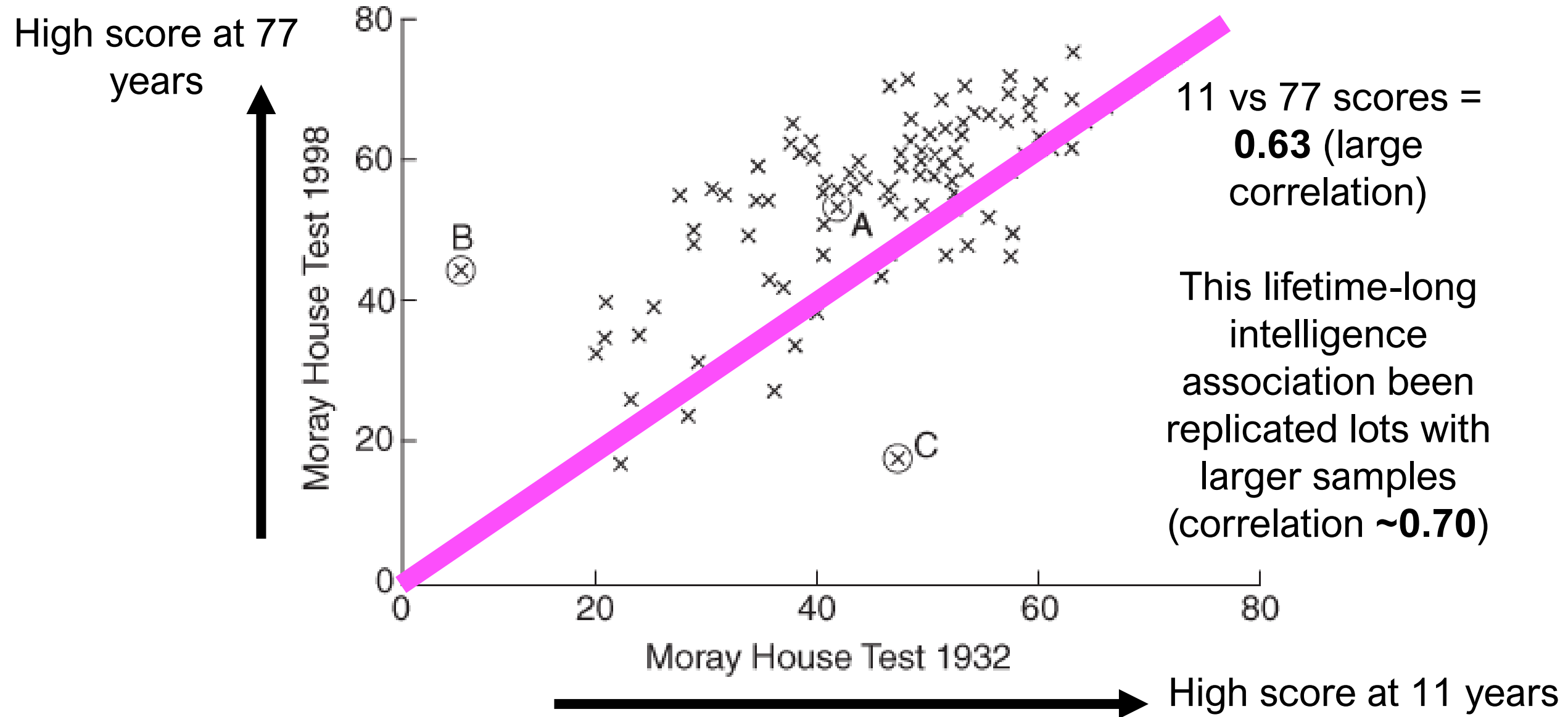
Intelligence at 11 vs 77 years of age



Intelligence at 11 vs 77 years of age



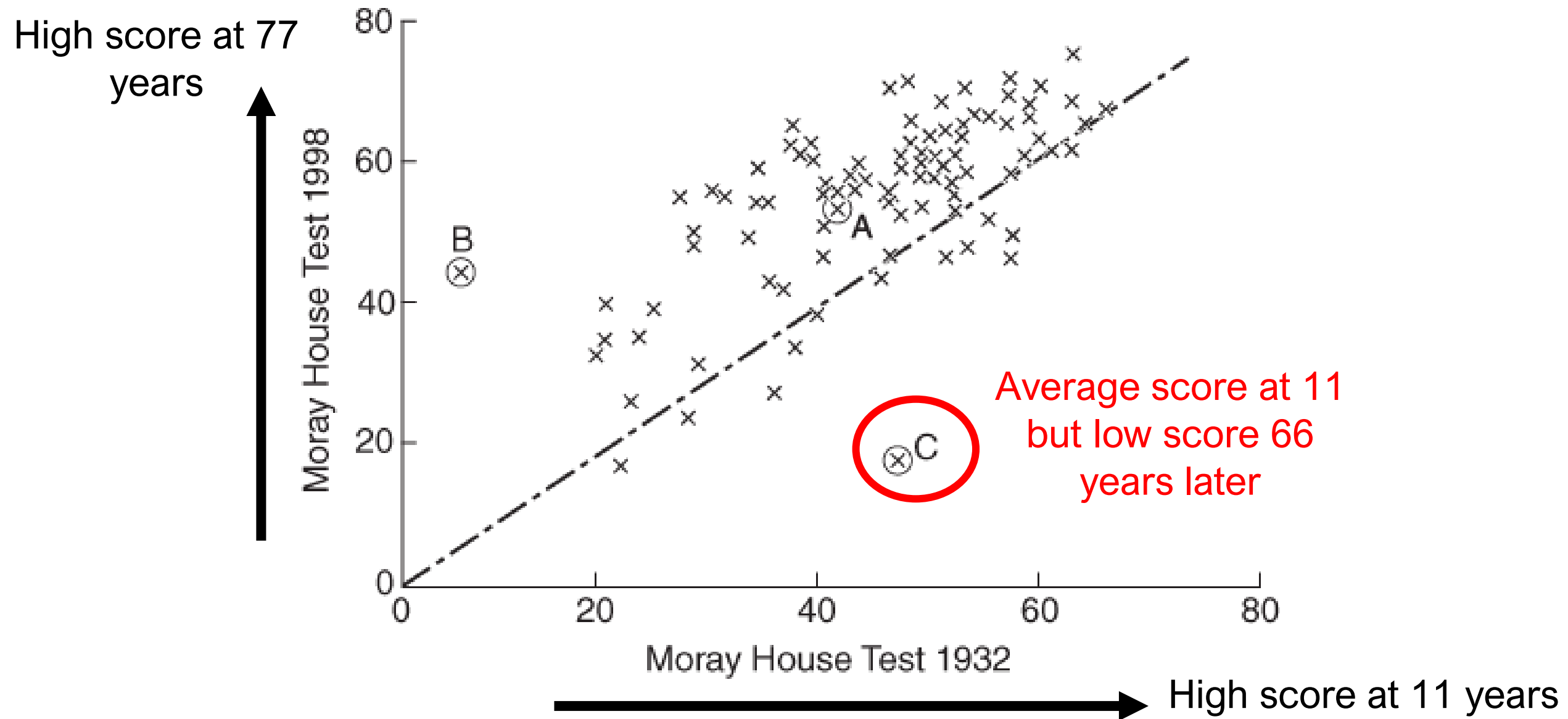
Intelligence at 11 vs 77 years of age



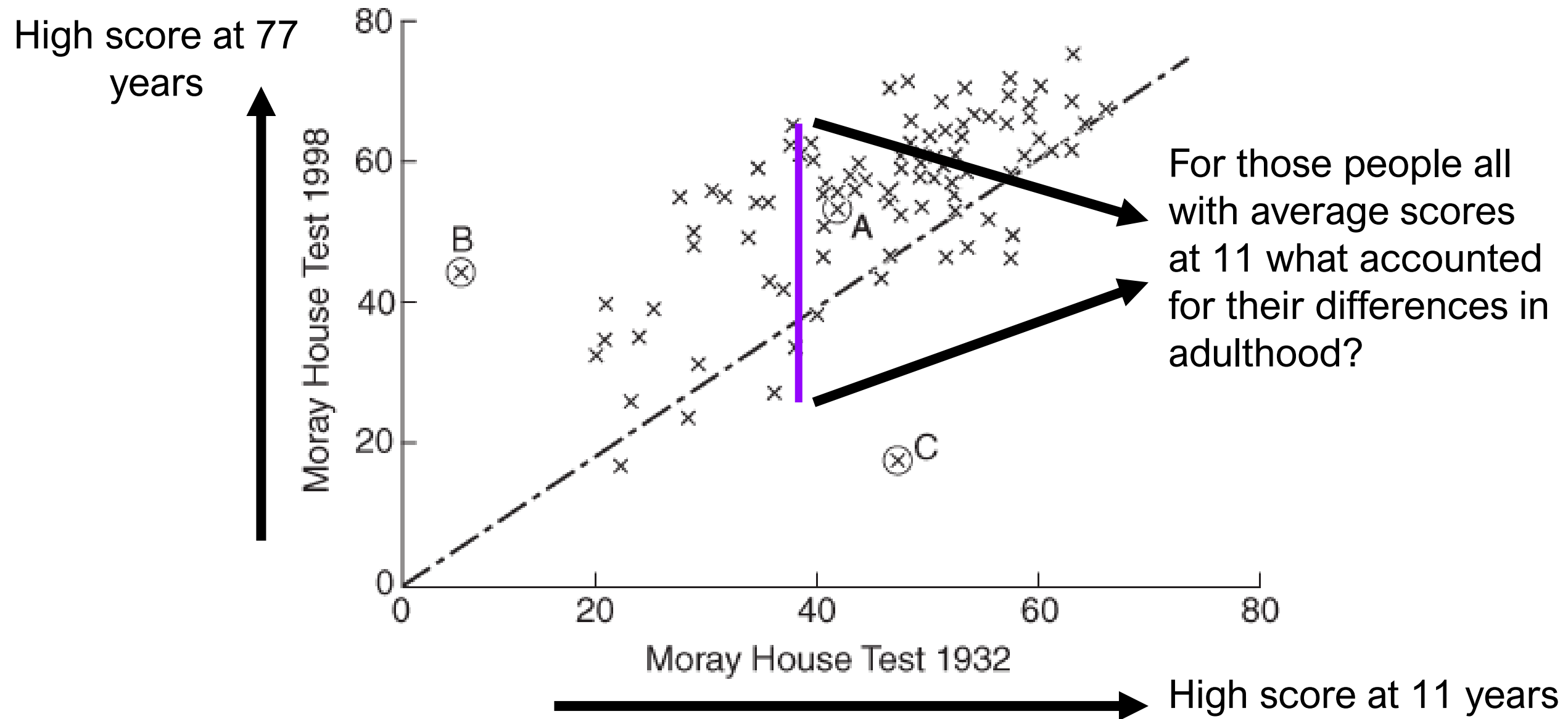
Intelligence at 11 vs 77 years of age



Intelligence at 11 vs 77 years of age



Intelligence at 11 vs 77 years of age



Summary of Scottish Mental Survey Finding

*After studying the Lothian cohorts' test scores on dozens of cognitive tests, sampling their genes, scanning their brains, and documenting their lifestyles and health in painstaking detail for more than 10 years, **Deary has found one factor that appears to predict late-life cognitive ability better than any other single measure. not exercise, education, or any other virtuous activity, but rather simply an individual's level of intelligence at age 11.***

As Deary likes to say about old age, quoting Fred Astaire, "To make a success of it, you've got to start young."

Why does some people's intelligence age better than others?

- Very difficult to answer as randomised control trials (ppl. randomly allocated to treatment groups) not easy across life-span
- Instead, most studies are observational (offer correlations between risk/protective factors and decline in intelligence)

Why does some people's intelligence age better than others?



Brenda Plassman

- Systematic review – found 150ish studies looking for risks or protections against cognitive decline
- Found some evidence of **risks**:
 - > APOE gene = higher cognitive decline
 - > Smoking
 - > Depression
 - > Diabetes

Why does some people's intelligence age better than others?



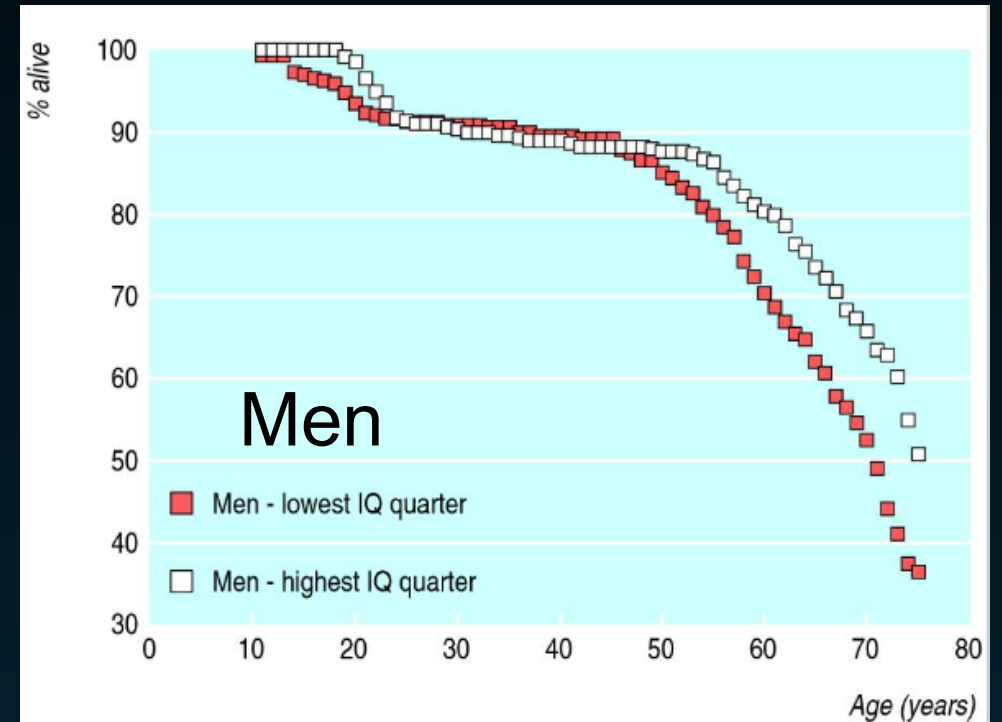
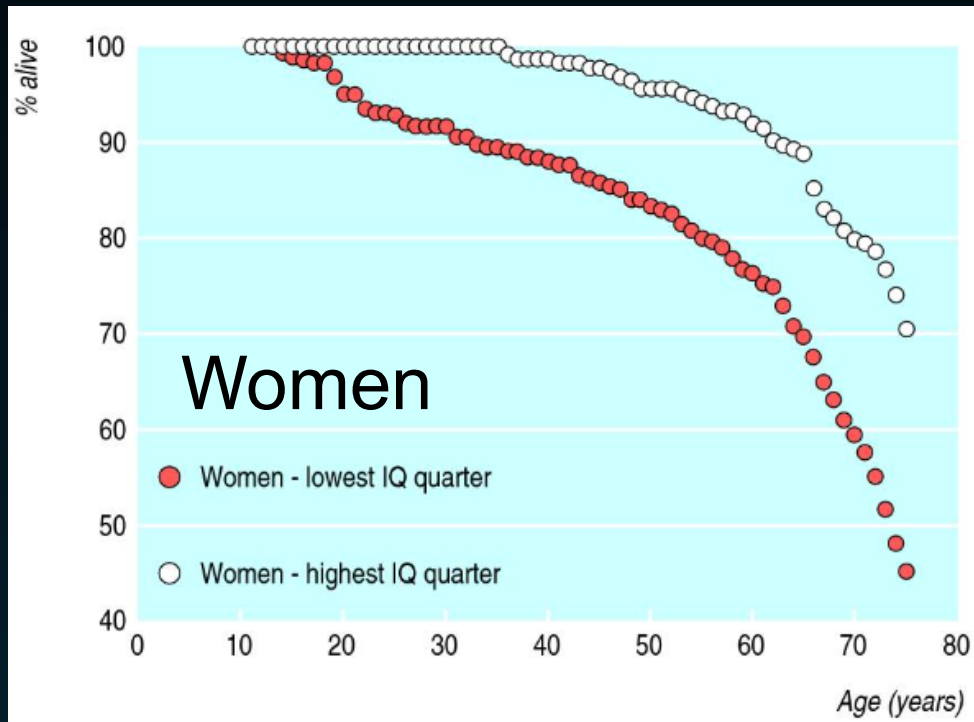
Brenda Plassman

- Systematic review – found 150ish studies looking for risks or protections against cognitive decline
- Found some evidence of **protective factors**:
 - > Mediterranean diet
 - > Eating more vegetable
 - > Being in more professional occupations
 - > Taking part in some leisure activities

Link between childhood intelligence scores and longevity

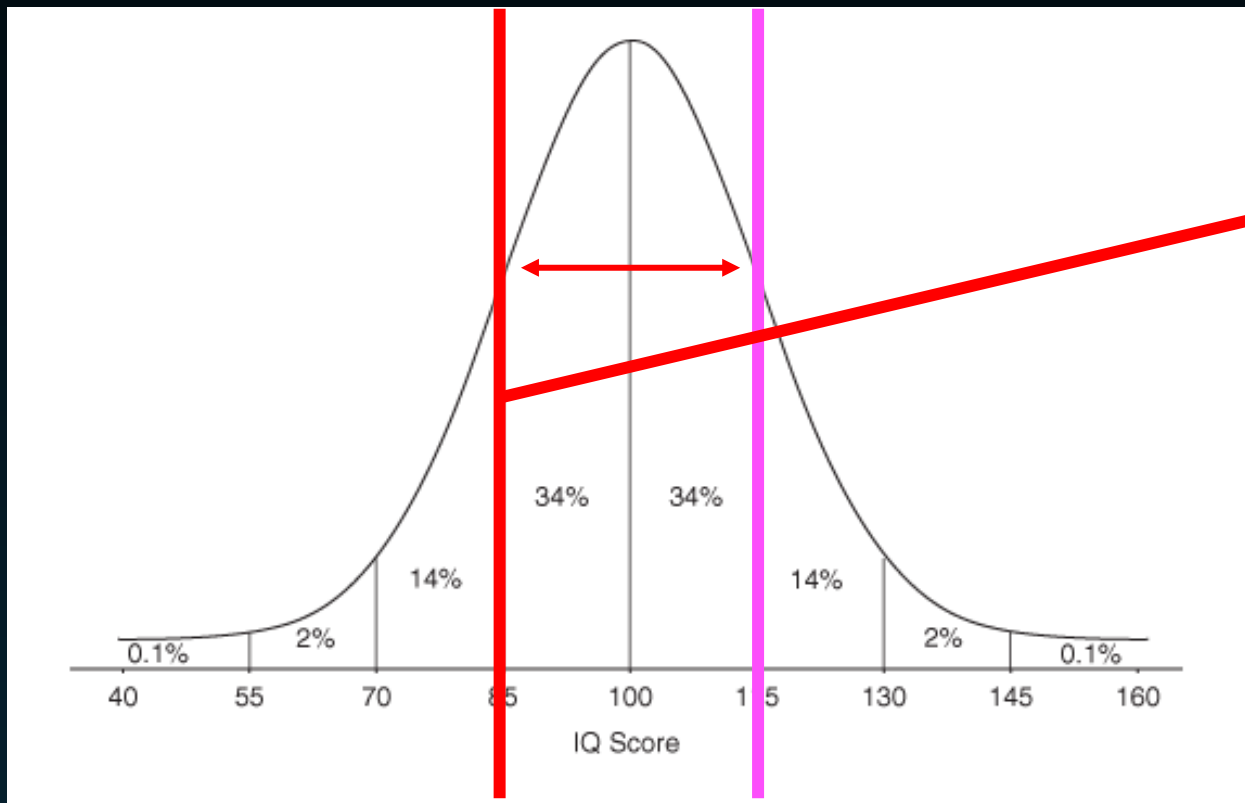
Scottish Mental Survey 1932 – Intelligence and longevity

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- Basically, looked to see if they were dead or alive up to 76 years



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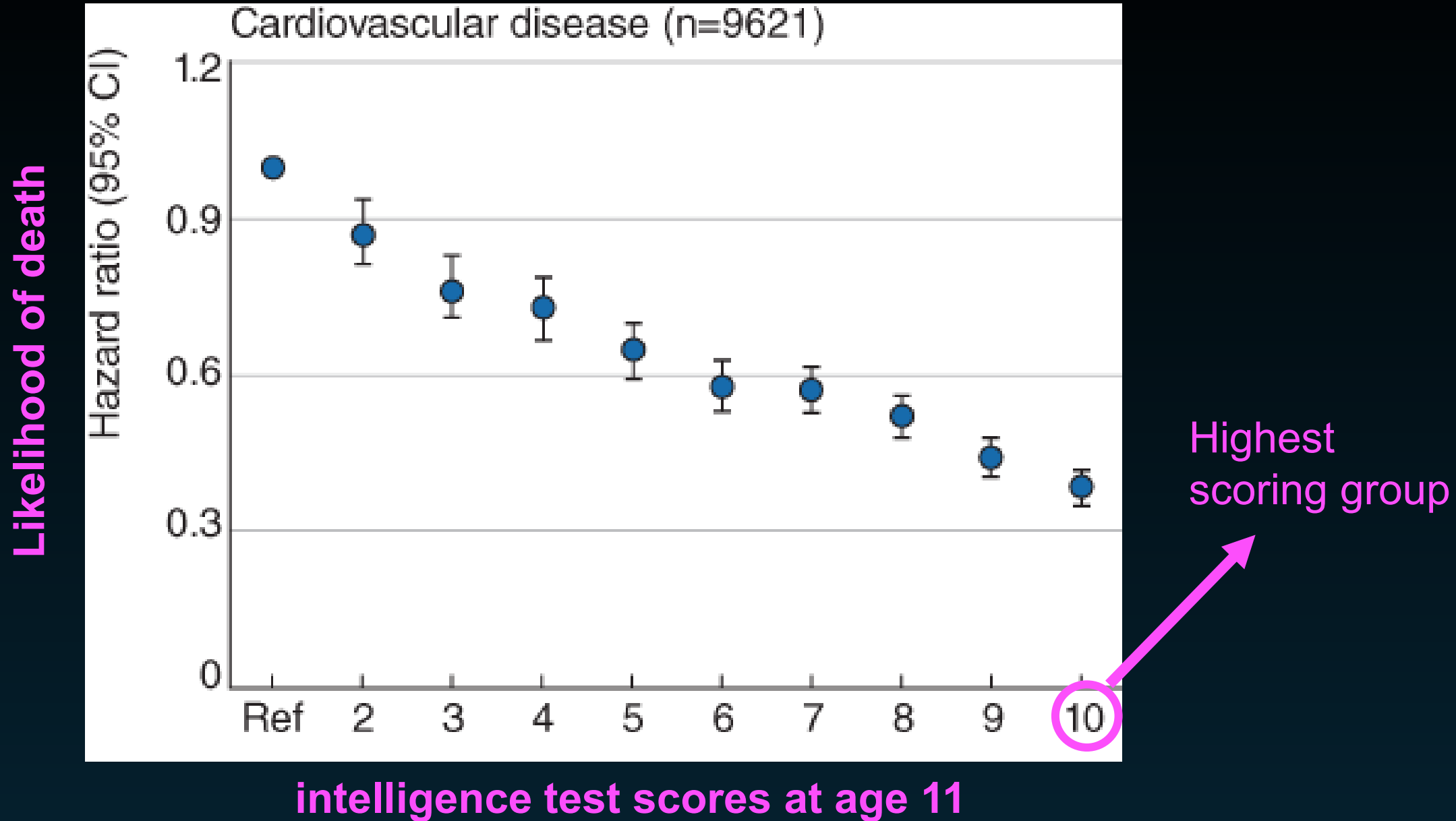


On average a girl with a 30 point disadvantage in IQ at age 11 was half as likely to be alive

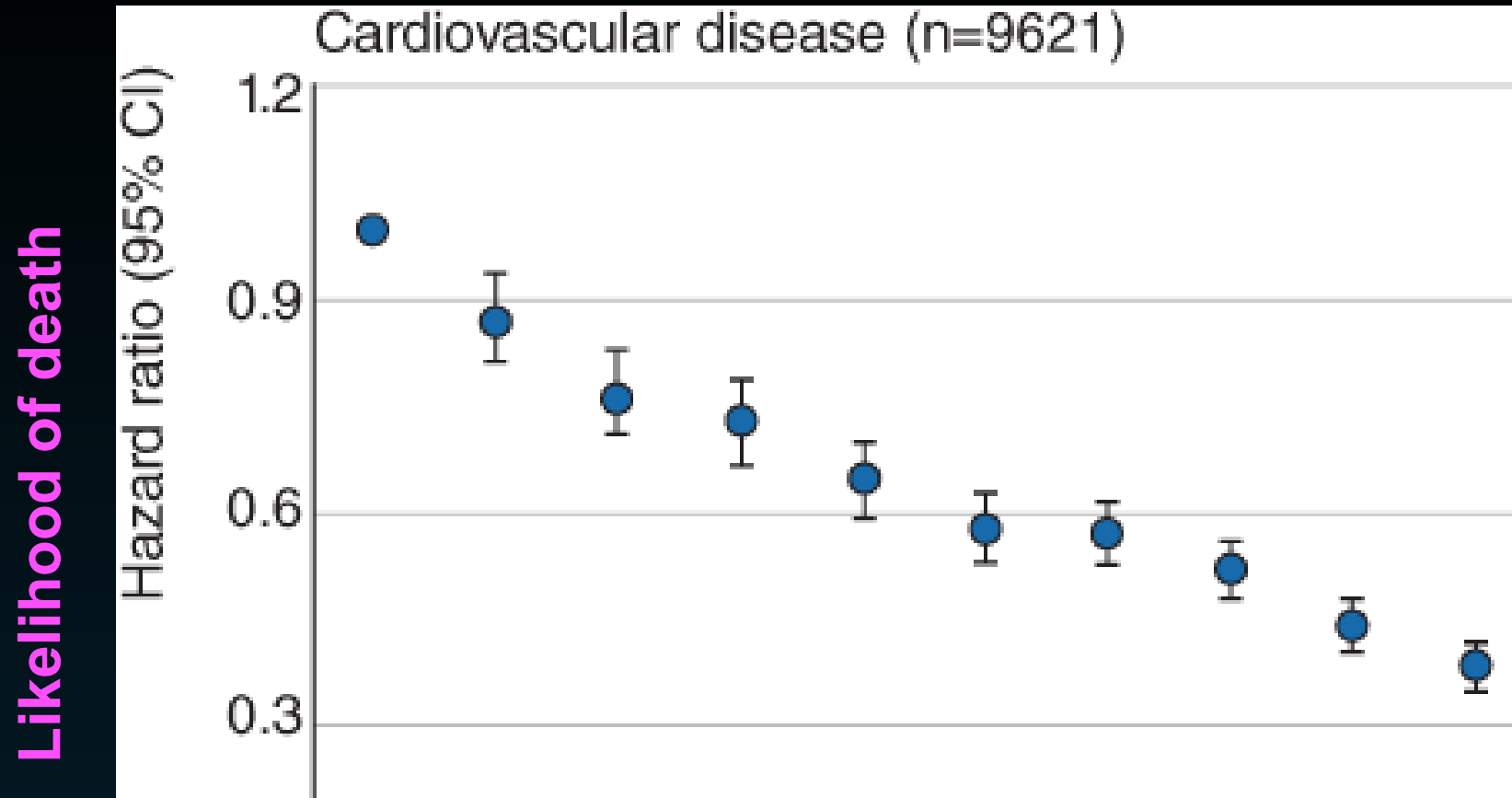
Scottish Mental Survey 1947

- Of the 70,805 children tested at 11 in 1947, 65,765 were traced via health records (92% of original survey) as of Dec 2015
- Those alive were 79 years old
- Average time to follow up was 57 years (25,979 had died, and 30,464 were still alive, 9,322 not known)

Scottish Mental Survey 1947



Scottish Mental Survey 1947



As the intelligence test score at age 11 becomes higher and higher, the risk of death from this cause by age 79 steadily lowers.

- Large meta-analysis of 16 studies across the world has replicated this finding, i.e. higher intelligence scores in youth (Calvin et al 2011), i.e. 15 IQ points in youth was associated with a 24 per cent lower risk of death during the follow-up periods.
- Adjusting for education and adult socio-economic status reduced the intelligence–mortality associations between a third and a half.
- However, there is a possibility that education and adult socio-economic status might reflect, to an extent, earlier intelligence level.

Link between youth intelligence and later health

The National Longitudinal Survey of Youth 1979

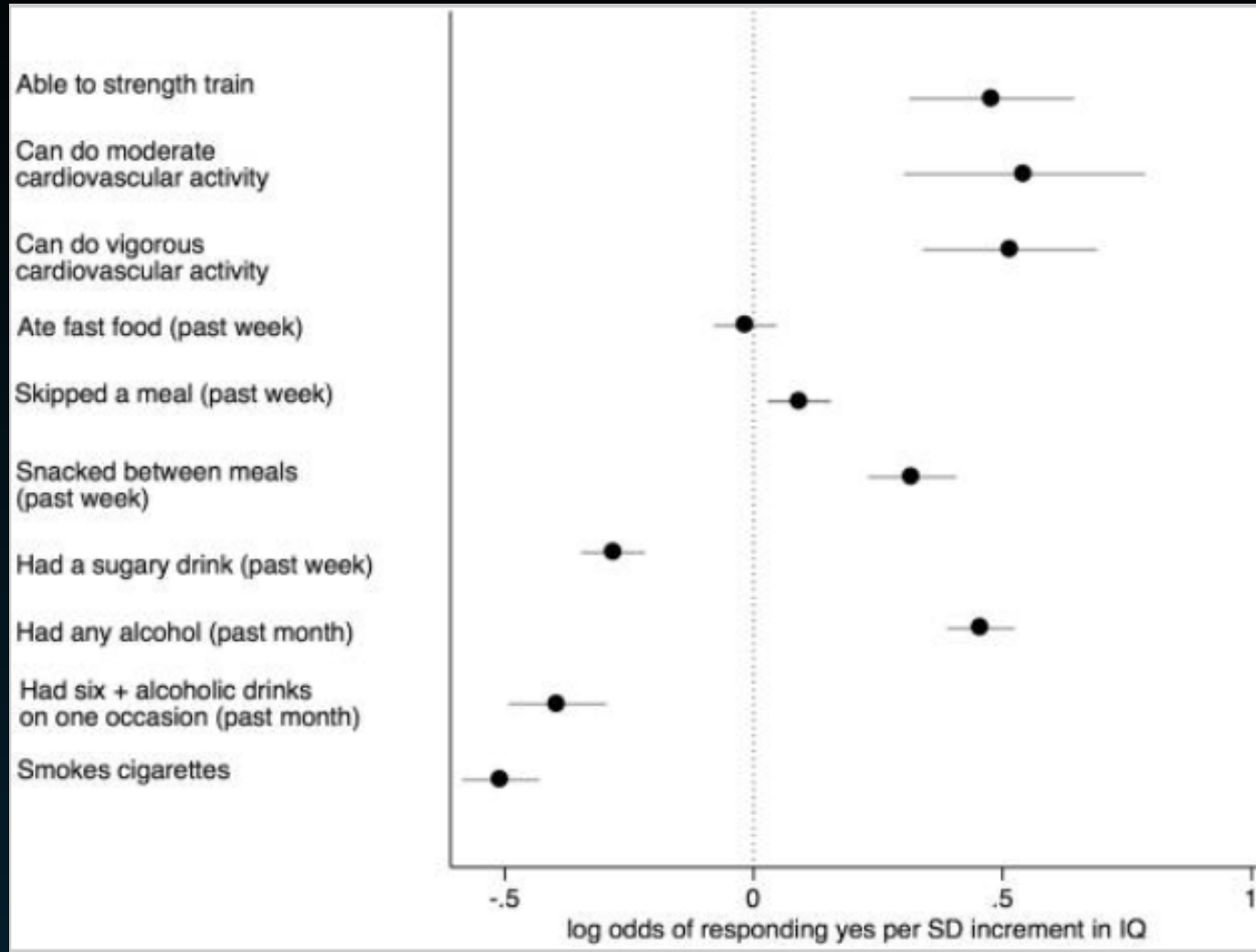
Christina Wraw and team reported that higher intelligence test scores in youth (14-21 years) acted as a protective factor for poor health later in life (~50 years old)

Percentage lower risk of a diagnosis per 15 IQ points advantage in the intelligence test score.

Thus, higher intelligence in youth was associated with lower risk of being diagnosed by age 50 with:

- High blood pressure (20%)
- Diabetes (15%)
- Lung Disease (29%)
- Congestive Heart failure (34%)
- Stroke (35%)

The National Longitudinal Survey of Youth 1979



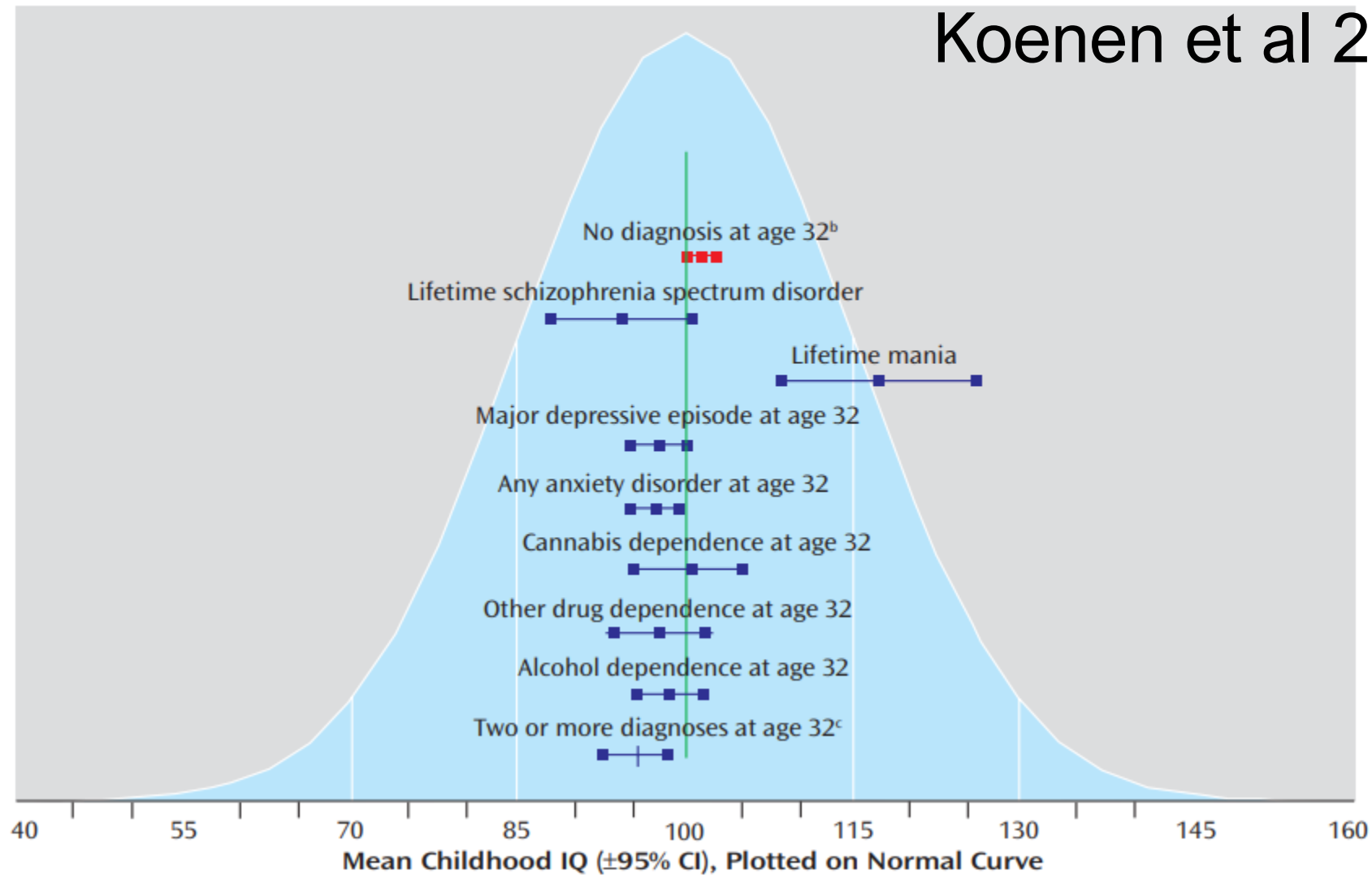
Also reported that those with 15 IQ point youth advantage less likely at 50 to smoke, drink heavily but more likely to workout etc.

Good study in that there is a pre-morbid measure of intelligence taken in youth across a large sample

Adjusting for childhood socio-economic status had little influence on these results. Therefore, childhood deprivation was not an explanation for the intelligence–health associations.

FIGURE 1. Childhood IQs for Individuals With Specific Mental Disorders at Age 32 in a Representative 1972–1973 Birth Cohort (N=935)^a

Koenen et al 2009



Lower childhood IQ predicts adult mental health. Why?

Intelligence test scores at childhood already hint at departure from optimum health

Potential Reasons for link between intelligence and health / longevity

Intelligence test scores at childhood might represent better wired body overall, from birth or even earlier 'system integrity'

Higher childhood intelligence leads to better education, jobs, and might have health advantages including being in safer environments

possibility that intelligence and health outcomes—including longevity—might have partially shared genetic contributions.

Higher childhood intelligence associated with uptake of healthier behaviours and lifestyles

Quick Summary

- There is a robust association between higher intelligence test scores in early life and living longer, and having a lower risk of dying from several causes.
- Looked at five possible and non-exclusive causes for the link between childhood intelligence and health and longevity.
- While socio economic status, education etc. can explain some of the variance, intelligence scores in youth appear predictive

END